

COSC2999 – Practical Data Science with Python

Assignment 3: Group Project

Prediction of Online Shoppers Purchasing Intention

Team members:

Vo Ngoc Dung - s4124370@rmit.edu.vn

Tran Nguyen Gia Phuc - s3891900@rmit.edu.vn

To Hung Phuoc – s3878094@rmit.edu.vn

Course Coordinator:

Dr. Thuy Nguyen

RMIT University Vietnam

Sai Gon South Campus

Date of Submission: 18th January 2025

***We hereby declare that this report represents my
own work, and all sources of information have
been appropriately acknowledged.***

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EXECUTIVE SUMMARY

In today's world of online commerce, businesses with web-based stores are constantly seeking ways to enhance the customer experience, aiming to boost sales. This enhancement can be achieved through various strategies, but they all begin with a fundamental question: identifying which customers are most and least likely to make a purchase. This report conducts in-depth customer behavior's analysis and utilizes machine learning to predict potential online buyers, leveraging the Online Shoppers Purchasing Intention Dataset from the UCI Machine Learning Repository [1]. The primary objective is to identify key factors leading to purchasing intention, segmenting customers and equipping businesses with insights to refine marketing strategies and improve customer engagement, thereby increasing revenue.

INTRODUCTION

In recent years, the online market, or e-commerce, has become increasingly popular and widespread in various kinds of products and countries. In Latin America, the number of online users rose 22.6%, from 126.8 million people in 2016 to 155.5 million in 2019, helping to increase the overall the profit into 79.9 billion U.S. dollars [2]. In other countries, the e-commerce market also experienced the same scenarios, becoming the highly earned profit in retailing merchandise [3]. Consequently, it is extremely crucial to investigate and study the behaviors of the online purchasing intentions of customers, so that the potential financial income can be earned at the most optimal way in the future. By exploring the chosen data set, meaningful insights can be gained, which would lay the solid foundation for attracting the users to buy new products.

The selected dataset for this project focuses on the e-commerce market with 12,330 feature vectors in total, consisting of 84% rows which are negative customers (who do not make purchase) and the rest are who make purchase [1]. Each row is the session of different users recorded in one year, which is used to prevent from recording any tendency to the unique events (specific campaign, special days, holiday periods, etc.). There are eighteen columns, including the target column Revenue. Ten columns are numerical type, and the rest are categorical values. In the first glance, the attributes are described in the two tables below [1].

Attribute Name (Numerical)	Attribute Description	Type of Variable	Min-Max Values
Administrative	Number of 'administrative' pages viewed	Numerical (Discrete)	Min is 0 and max is 27
Administrative_Duration	Total amount of time (in seconds) spent by the visitor on 'administrative' pages	Numerical (Continuous)	Min is 0 and max is 3,398.75
Informational	Number of 'informational' pages viewed	Numerical (Discrete)	Min is 0 and max is 24
Informational_Duration	Total amount of time (in seconds) spent by the visitor on 'informational' pages	Numerical (Continuous)	Min is 0 and max is 2,549.37
ProductRelated	Number of 'product related' pages viewed	Numerical (Discrete)	Min is 0 and max is 705
ProductRelated_Duration	Total amount of time (in seconds) spent by the visitor on 'product related' pages	Numerical (Continuous)	Min is 0 and max is 63,973.5
BounceRates	Average bounce rate value of visitors who enter the site from that page and then leave ("bounce") without triggering any other requests to the analytics server during that session	Quantitative, Continuous	Min is 0 and max is 0.2
ExitRates	Average exit rate value of the pages visited by the visitor	Quantitative, Continuous	Min is 0 and max is 0.2
PageValues	Average page value of the pages visited by the visitor	Quantitative, Continuous	Min is 0 and max is 361
SpecialDay	Closeness of the site visiting time to a special day (e.g. Mother's Day, Valentine's Day) in which the sessions are more likely to be finalized with transaction	Quantitative, Continuous	Min is 0 and max is 1

Table 1: Attribute description - Numerical Type

Attribute Name (Categorical)	Attribute Description	Type of Variable	Min-Max Values
Month	Month value of the visit date (Feb, Mar, May, June, Jul, Aug, Sep, Oct, Nov, Dec)	Ordinal	Listed in the attribute description
OperatingSystems	Operating system used for the session	Nominal	From 1 to 8
Browser	Browser used for the session	Nominal	From 1 to 13
Region	Geographic region from which the session has been started by the visitor	Nominal	From 1 to 9
TrafficType	Traffic source by which the visitor has arrived at the Web site (e.g., banner, SMS, direct)	Nominal	From 1 to 20
VisitorType	Type of visitors (New Vistor, Returning Visitor, and Other)	Nominal	Listed in the attribute description
Weekend	Whether the session occured on weekend	Binary	True, False
Revenue	When a user's online session results in a completed purchase the value is True otherwise False	Binary	True, False

Table 2: Attribute description - Categorical Type

The research question revolves around:

1. Classification – How to accurately identify potential customers based on their online behaviors?

For the classification models, this notebook will employ Random Forest Classification and Stacking Model to tackle the binary-class classification problem, measuring the user's intention to finalize the transaction.

2. Clustering – What are the common online behaviors or patterns that potential customers would have?

For the clustering models, this notebook willll apply K-Means and DBSCAN models to segment users based on 4 main features ***ProductRelated, ProductRelated_Duration, BounceRates, ExitRates***.

By setting out two questions which can be seen as the achieved goals, each step in this data science project will be clearly conducted and executed appropriately.

METHODOLOGY

In this project, we adopted a standard data science workflow using Python alongside essential libraries, including Scikit-Learn, Pandas, Numpy, and Seaborn, to facilitate various data preprocessing, analysis, and modeling tasks. **The initial step** involved preprocessing a web-based UCI dataset to ensure data integrity. This included removing null and duplicate rows, standardizing data types, and validating value consistency, outliers removal, all crucial for preparing a reliable dataset for analysis.

Following preprocessing, we conducted Exploratory Data Analysis (EDA) to uncover meaningful patterns and relationships within the data. Through visualizations such as box plots, scatter plots, histograms, and bar charts, combined with descriptive statistics like mean, standard deviation, and range, we gained insights that guided our feature selection process for the subsequent modeling phase.

In the feature engineering stage, we focused on identifying the most relevant variables for predictive modeling. Categorical features were one-hot encoded, while the target variable, "Revenue," was label-encoded. We refined the feature set through scaling techniques and the insights garnered from EDA. Multiple methods, including Correlation Matrix analysis, Mutual Information with SelectKBest, Chi-square tests, Univariate ROC-AUC, and Random Forest Recursive Feature Elimination (RFE), were employed to ensure we selected features that were consistently ranked highly, laying the foundation for meaningful predictions.

Recognizing the target variable's imbalanced nature (85% non-purchase vs. 15% purchase), we implemented a combination of undersampling and oversampling techniques to balance the class distribution effectively. After addressing class imbalance, we developed two types of classification models, Random Forest and Stacking, alongside two clustering models, K-Means and DBSCAN. The classification models aimed to predict purchase likelihood based on user activities, while the clustering models sought to identify behavioral patterns among purchasing customers.

Then, we trained the models using default parameters and evaluated their performance. To further enhance predictive accuracy and model effectiveness, we performed hyperparameter tuning using GridSearchCV and RandomizedSearchCV. This step was crucial to optimize model parameters while balancing complexity and mitigating risks of overfitting or underfitting.

Finally, we utilized varied evaluation metrics tailored to the specific tasks at hand. For classification models, we implemented metrics such as the Confusion Matrix, Classification Report, ROC-AUC, and Precision-Recall Curve, alongside Cross-Validation techniques, to assess performance. In contrast, the quality of clustering models was evaluated using the Silhouette Score. By comparing and contrasting these metrics, we developed a comprehensive understanding of each model's effectiveness and overall performance.

RESULTS

1. Categorical Features EDA

In this section, we examine the impact of each of the eight categorical features on the class label from two perspectives: "frequency" and "probability." Due to space constraints, we will only discuss a selection of these features in this paper.

Month

Figure 1 illustrates how revenue generation is influenced by different months. The bar chart on the left, labeled "frequency," highlights that certain months (March, May, November, and December) show higher activity levels compared to others. However, it does not clearly indicate whether these months have a greater likelihood of purchases. To address this, a "probability" chart was created, shown on the right side of Figure 1. This chart reveals that while the overall impact of the month on revenue generation is minimal (with the highest probability barely exceeding 20%), some months, notably October and November, exhibit a slightly higher probability of purchases compared to others.

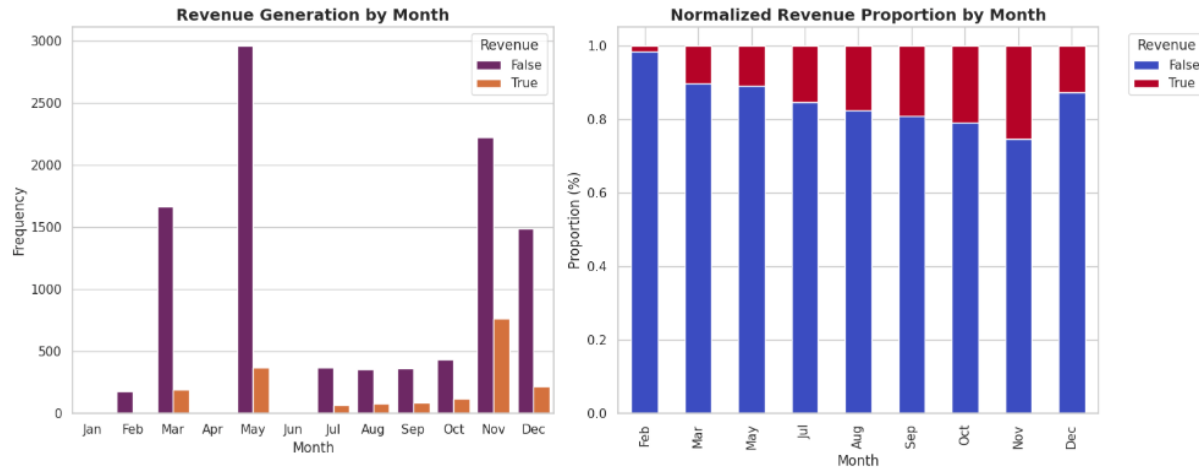


Figure 1: Revenue on each Month

Operating System (OS)

Figure 2 illustrates how the operating system influences revenue generation. The "frequency" bar chart on the left reveals that most customers use only a few specific operating systems. However, as with the earlier "month" chart, it is challenging to determine whether any particular OS significantly affects the probability of a purchase. The normalized chart on the right indicates that the operating system appears to have minimal impact on revenue generation.

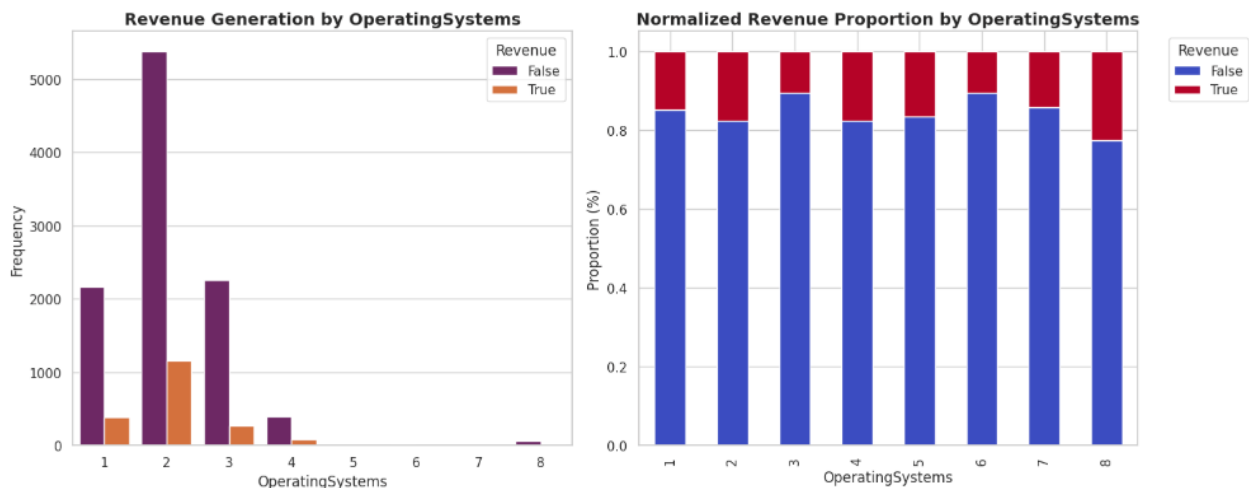


Figure 2: Revenue on each Operating System

Visitor Type

The bar chart depicts revenue generation across three categories of visitors. Approximately 13.9% are new visitors, 85.5% are returning customers, and a small fraction, around 0.7%, falls into the "other" category. The visualization demonstrates that returning visitors contribute the most to website activity and generate the highest number of purchases compared to the other groups.

When viewed from a "probability" perspective, as illustrated in Figure 3, new visitors show a slightly higher likelihood of making a purchase than the other two types of visitors, although the difference is minimal.

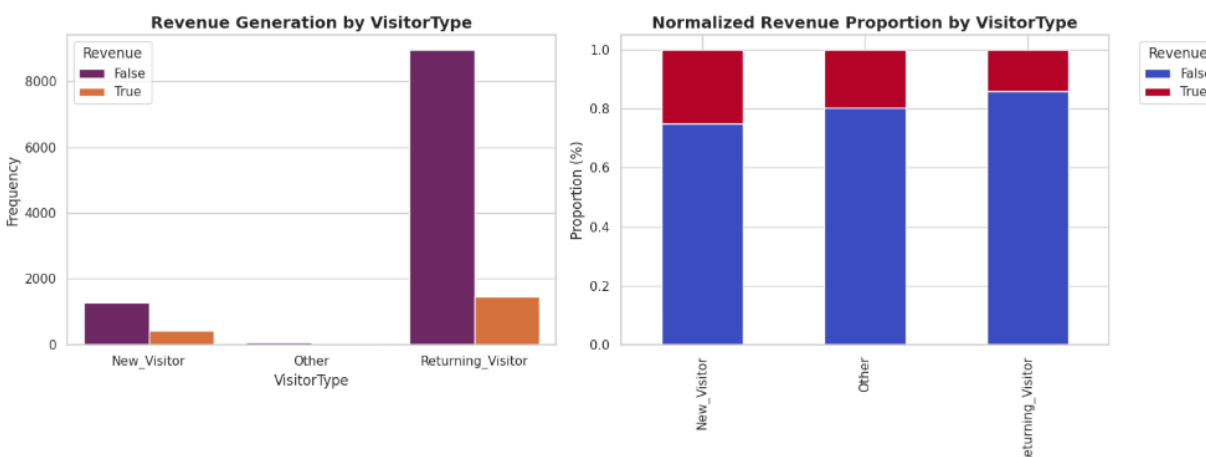


Figure 3: Revenue on Each Visitor Type

Region

The examination of the region variable offers important insights. The figure below highlights the geographical locations from which the website was accessed. It reveals that the majority of visits originate from regions 1, 2, 3, and 4, while regions 5 through 9 account for a much smaller proportion of visits, as shown below. However, the "probability" chart in Figure 4 indicates that the likelihood of revenue generation is fairly consistent across all regions, with no significant differences observed.

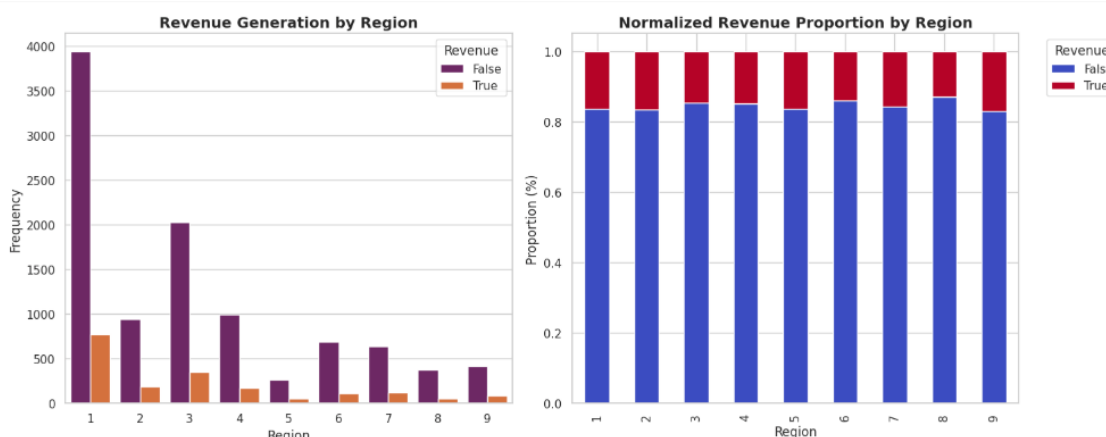


Figure 4: Revenue on each Region

2. Numerical Features EDA

We analyzed the effect of the ten numerical features on the target variable using a mix of "frequency" and "probability" histograms, along with occasional scatterplots. However, due to space constraints, this report focuses on discussing only a select few of these features.

Visit Pages

According to the data, customers accessed the ProductRelated webpage more frequently than the Administrative or Informational pages. Based on this observation, the Administrative page appears to function as the website's homepage, while the Informational page serves as the "Contact Us" section, and the ProductRelated page is where detailed product information is presented. However, this contextual information was not originally provided in the dataset.

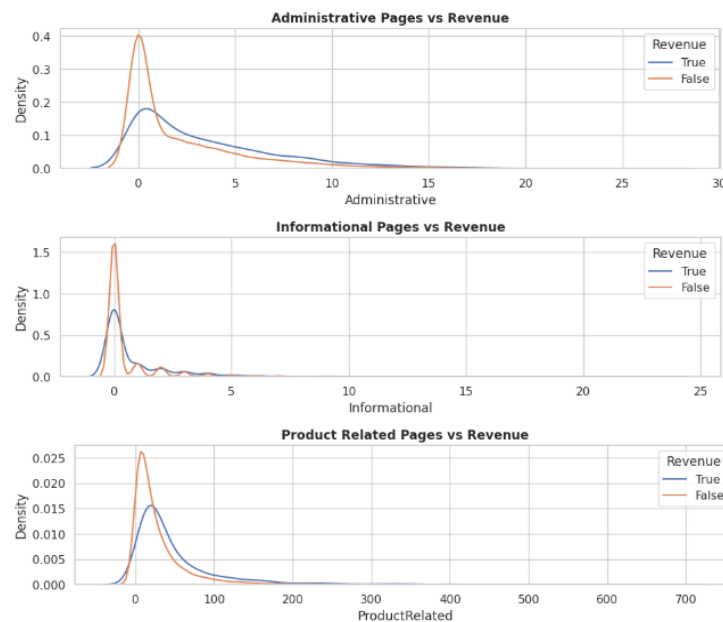


Figure 5: Revenue on each Page type

Bounce Rate has an inverse correlation with revenue creation. The less the bounce rate values are, the more chances customers will buy the products.

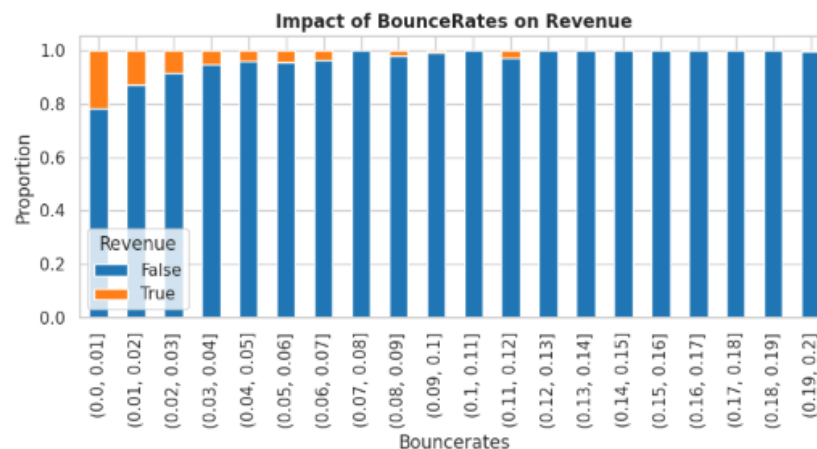


Figure 6: Impact of Bounce Rate on Revenue

Page Values: The density peak not buying products is much sharper near zero compared to make purchasing, indicating that non-purchasing users tend to interact with low-value pages. A higher PageValues appears to correlate positively with Revenue, implying that users engaging with higher-value pages are more likely to convert.

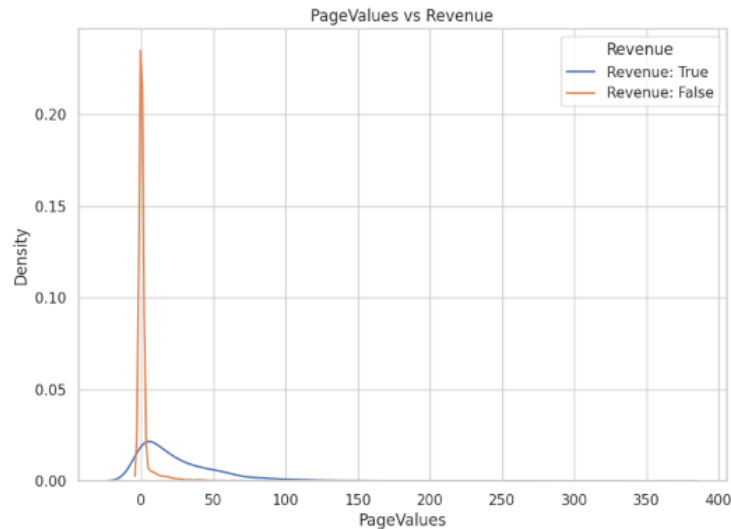


Figure 7: Impact of Page Values on Revenue

Exist Rates: A lower exit rate is strongly correlated with a higher likelihood of generating income.

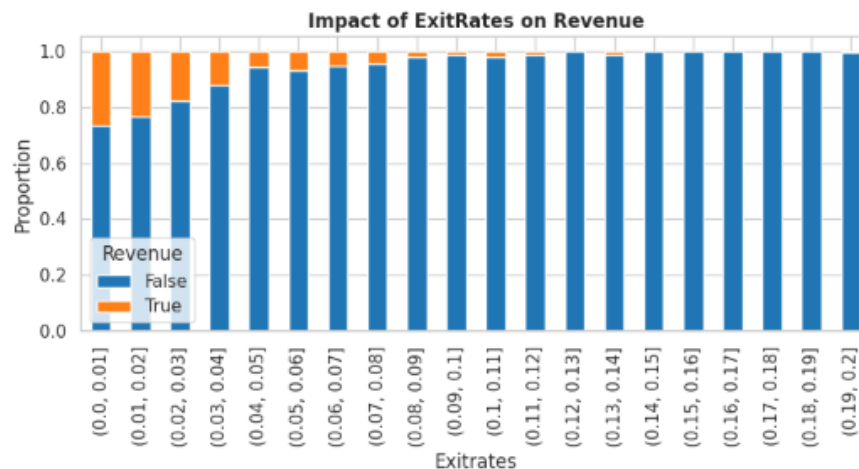


Figure 8: Impact of ExitRate on Revenue

Special Days: Most interactions or sessions occurred on non-special days. The data does not indicate a relationship between special days and revenue generation. This may be assumed as a notable finding, as people might expect higher revenue on special days, but the data does not support that hypothesis.

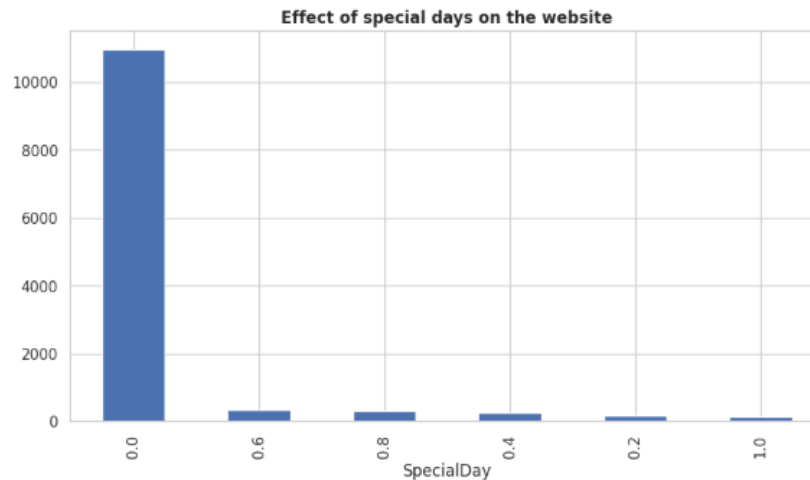


Figure 9: Effect of special days on revenue

Bounce Rates vs. Exist Rates

Hypothesis: Higher bounce rates are expected to be positively correlated with higher exit rates in e-commerce, as both metrics reflect users leaving the site prematurely. Pages with high bounce rates likely indicate that users are not finding the desired information or content, leading to a higher likelihood of exiting the site altogether.

Intuition: The scatterplot reveals a strong positive correlation between bounce rates and exit rates, as evidenced by the upward-sloping regression line. This trend indicates that as bounce rates increase, exit rates also tend to rise. Pages with low bounce rates exhibit relatively low exit rates, suggesting that these pages may successfully engage users, encouraging them to explore further. Conversely, high bounce rates correlate with higher exit rates, pointing to potential issues such as poor user experience, irrelevant content, or technical problems. This insight emphasizes the need to optimize pages with high bounce rates to improve user retention and reduce exits.

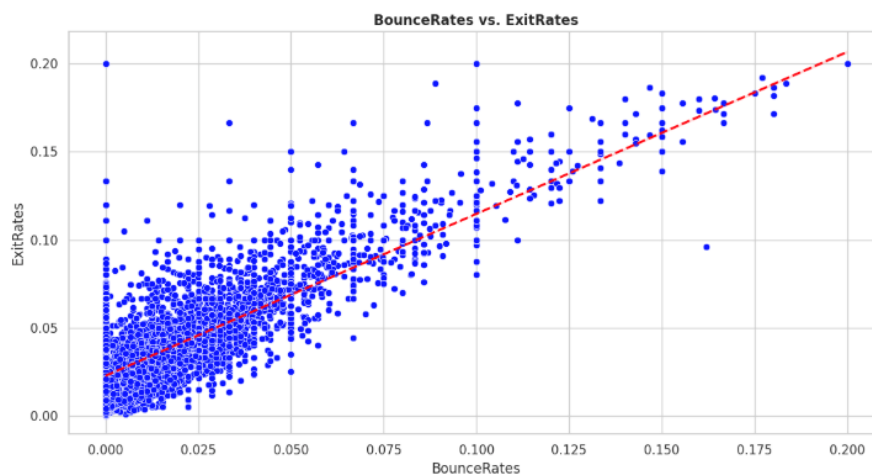


Figure 10: Positive Correlation between BounceRate and ExitRate

The Pairwise Plot

Intuition: The pairwise plot highlights relationships between various attributes and the Revenue as follows:

- There is a positive correlation between the number of ProductRelated pages visited and ProductRelated_Duration, in which users with higher engagement in these pages are more likely to complete a transaction.
- Lower BounceRates and ExitRates are associated with completed transactions, suggesting better engagement with the website.
- Administrative and informational pages have minimal influence on revenue, suggesting a focus on optimizing product pages and engagement metrics.
- Higher PageValues are strongly associated with completed transactions, this metric could serve as a predictive factor, indicating the likelihood of purchase.
- A negative relationship exists between BounceRates and PageValues. Higher PageValues are associated with lower BounceRates, which is intuitive since engaging content reduces user drop-off rates. For transactions (Revenue=1), this pattern is particularly prominent, suggesting that bounce rates are a key metric for identifying and improving conversion potential.
- A positive relationship between the number of ProductRelated pages visited and SpecialDay indicates that users tend to explore more products during special promotional periods.

Summary of EDA Findings

The univariate analysis provided valuable insights into the individual characteristics of online shoppers, highlighting their browsing behaviors, visit frequency, and shopping tendencies. The bivariate analysis identified key relationships between variables, such as visitor type, session duration, and revenue generation, suggesting potential factors contributing to purchase decisions.

DISCUSSION

After analyzing the relationships between various features, several key insights were derived:

- Customers tend to visit the Product_Related pages more frequently than other sections of the website. Administrative pages likely serve as an introduction to the shop or company, while Informational pages provide contact details. In contrast, Product_Related pages focus on product information, reflecting their importance in the shopping process.
- Although March, May, and December see higher overall activity, October and November stand out for their significantly higher probability of purchases. This trend likely aligns with the holiday shopping season, as consumers are more inclined to make purchases for gifts and personal use during this period, suggesting a seasonal buying pattern.
- While minor differences in revenue generation were observed across regions, traffic types, days of the week, and operating systems, these variations are not particularly significant.

- Browser one and Browser two outperform others in revenue generation, likely due to their user-friendly interfaces and attractive designs. Other browsers contribute minimally to revenue, underscoring the importance of optimizing for top-performing platforms.
- New visitors exhibit the highest likelihood of making purchases, emphasizing the need to focus on consistently attracting and engaging this group. Converting new visitors into loyal customers could be a powerful strategy for driving revenue growth.
- Finally, maintaining a low bounce rate, reducing exit rates, and increasing page values are essential for boosting revenue. By keeping customers engaged and encouraging them to explore more pages, businesses can enhance the likelihood of conversions, leading to higher sales and sustained growth.

For categorical columns, one-hot encoding is used, while label encoding is applied to the target revenue column. Numeric columns undergo Standard Z-score scaling to prevent overfitting in scale-sensitive models like Support Vector Machine. After preprocessing, we have fifty-seven features, necessitating feature selection to identify the best ones for model training.

Feature Selection

In the classification task, feature selection enables ML algorithm to train faster, reduces the complexity of the model, reduces overfitting, and increases accuracy. To maintain a comprehensive and strategic approach, we utilize a variety of techniques to identify the most relevant features. Our methodology involves selecting features that consistently prove significant across multiple selection techniques. While this approach is meticulous and exhaustive, we anticipate it will yield superior results. Below, we outline the techniques employed for feature selection, as well as the features identified as most critical according to each method:

- **Correlation Exploration:** After using Brute Force approach, there are five features with highly correlated values, which is over predefined threshold value of 0.8. To identify which features need to be removed, we continue examining them by plotting the table showing a group of correlated features. Finally, four features to delete are: **Browser_1**, **BounceRates**, **Weekend_True**, **VisitorType_Returning_Visitor**.
- **Mutal Information and SelectKBest:** top 25 best features are kept; other ones will be removed.
- **Chi Square:** The output obtained includes p-score values, indicating the associated dependence of those features and target columns. If the p-score value is low, that feature will be incredibly significant and contribute to final predictions of the models.
- **Univariate ROC_UAC:** The output obtained includes p-score values, indicating the associated dependence of those features and target columns. If the p-score value is low, that feature will be incredibly significant and contribute to final predictions of the models.
- **Random Forest Recursive Feature Elimination:** By default, the features whose importance score is greater than mean of all importance scores are kept for modelling.

The final variables that we select are the following. We have selected them based on respective frequency i.e. number of techniques in which they are important. There are 25 features in total that will be kept.

Meanwhile, four features, which are **ProductRelated**, **ProductRelated_Duration**, **BounceRates**, **ExitRates** are suitable to meet the requirements of the predefined goals for clustering tasks.

Handling imbalanced class using SMOTE

The Revenue column in the dataset is highly imbalanced, with 84% negative customers and only 16% positive customers, which can negatively impact model performance. To address this issue, it's essential to adjust the dataset using techniques like SMOTE (Synthetic Minority Over-sampling Technique) [5]. SMOTE enhances the representation of the minority class by generating new synthetic data points through interpolation between existing minority samples and their neighbors. This method increases the diversity of the minority class, improving the balance within the dataset and enhancing the model's predictive accuracy in binary classification tasks.

Model Training - Hyper Tuning - Evaluation

1. Random Forest

The Random Forest Classifier, trained with 200 estimators on a balanced dataset (using resampled training data), exhibited commendable performance. The model achieved a perfect training accuracy of 1.0, reflecting its ability to capture patterns in the training data. On the testing set, it attained an accuracy of 87.5%, indicating reliable generalization. The confusion matrix reveals that the model correctly identified 2,755 instances of the majority class ("Non-revenue") and 448 instances of the minority class ("Revenue"), with 335 and 124 misclassifications, respectively.

The ROC AUC score of 0.837 underscores the model's strong capability in distinguishing between classes. However, class-specific performance metrics indicate an imbalance: precision, recall, and F1 score for "Non-revenue" are 0.96, 0.89, and 0.92, respectively, while for "Revenue," these metrics drop to 0.57, 0.78, and 0.66.

Hyperparameter Tuning: After suitable hyperparameter tuning, the overall performance of Random Forest model does not improve much. All the evaluation metric values are not increased, which suggests that the model is not improved.

2. Stacking

The stacking model, or Stacked Generalization, is one of the ensemble learning models that combines outputs from multiple machine learning models. There are two levels in Stacking model. Level-0 models, or base models, are the set of multiple basic models that are used to train directly from training data set. Each model will produce the output and predictions, which are the input used to train the level-1 model. Level-1 model, or called Meta model, will depend on the outputs of level-0 to conclude the final outcome.

In this project, base models include Decision Tree, XGBoost (Extreme Gradient Boosting), Random Forest, and Meta Model is Logistic Regression model. Finally, the hyperparameter tuning, which is the cross-validation strategy with 10 Folds, is used to prevent the overfitting issues.

Overall, the performance of Stacking model is very impressive, and it even improves if combined with 10 folds cross-validation strategies. The achieved accuracy is 0.88, indicating the model predicts most of objects correctly. The confusion matrix reveals that the model correctly identified 2,760 instances of the majority class ("non-revenue") and 453 instances of the minority class ("Revenue"), with 330 and 119 misclassifications, respectively. However, class-specific performance metrics indicate an imbalance: precision, recall, and F1 score for "non-revenue" are 0.96, 0.89, and 0.92, respectively, while for "Revenue," these metrics drop to 0.58, 0.79, and 0.67. That may indicate that the model does not treat the imbalanced class properly and well enough as majority class. After cross-validation, the performance is improved significantly, with a mean accuracy of 100.0% ($\pm 0.0\%$), mean precision of 100.0% ($\pm 0.0\%$), and mean ROC AUC score of 100.0% ($\pm 0.0\%$). That means the stacking model becomes more robust and well predicts the unseen objects in the future.

3. K-Means Clustering

K-means clustering model is widely used and popular in unsupervised learning problems. The model will group the data points based on their similarity and characteristics of their features. Although K-means model is very easy to implement and computationally efficient, the requirement for K-means is the number of clusters need to be specified before using. Selecting the best value for k is highly crucial because it will effectively minimize the Sum of Squared Errors (SSE) while avoiding the unnecessary model complexity. In order to determine the optimal number of clusters (k), the hyperparameter tuning method, Elbow method, is used. The Elbow method would depend on the inertia values, which is Sum of Square Error (SSE) to find the best k. Finally, it is suggested that the number of cluster k is equal to 3.

After training the model, it is continued evaluating by The Silhouette Score. Silhouette score is a measurement assessing the quality of clustering, based on the distances of sample point to centroid of other clusters that do not include sample point. The score ranges from -1 to 1, with 1 indicates the clusters are well-grouped, and -1 indicates the points are grouped in wrong cluster. The received score for the K-Means clustering with number of clusters of 3 is 0.497, indicating slightly moderate cohesion and separation between the clusters. While this score suggests reasonable clustering quality, there's potential for improvement, such as adjusting the value of k or exploring alternative clustering algorithms.

There are three clusters, as indicated equal to k. **Cluster 0** represents engaged users with low BounceRates, low ExitRates, high interest in products (high ProductRelated), and high significant time spent on the site (high ProductRelated_Duration). This group represents likely active users with a high probability of conversion. Meanwhile, **Cluster 1** gathers users with varying behaviours, spending more time on the website but with moderate product interest (values close to 0). This group is likely seeking information but may not be ready to purchase. Finally, **Cluster 2** exhibits the highest BounceRates and ExitRates, representing "quick exit" users with minimal time spent on the site and limited product views. This group shows little to no interest in the website.

4. DBSCAN

DBSCAN (short form of Density-Based Spatial Clustering of Applications with Noise) is one of the most widely used models in unsupervised machine learning algorithm. One of its advantages of DBSCAN is that it does not require the pre-specified number of cluster like K-Means algorithm. Two parameters for DBSCAN are epsilon, maximum distance between two points in same clustering, and min_samples, the minimum samples required to form a new cluster. By successfully defining these two parameters, the model can achieve the optimal reliable output. To do that, Grid Search was used to determine the optimal DBSCAN parameters. The hyperparameter-tuning range is from 0.5 to 1.0 for epsilon, increasing 0.1 each turn, and integer value from 5 to 9 for range of minimum samples. The outcomes are an epsilon of 0.9 and a minimum samples value of 8. The Silhouette Score is again used to analyse the performance of DBSCAN, which produces a score of 0.742. That may imply that the clustering effectively identified meaningful customer segments.

DBSCAN identifies three clusters. Cluster (-1) represents noise or outliers, which indicates that the points do not fall into any groups of cluster. In this scenario, noise is represented as the unusual and extreme behaviours from online customers. Meanwhile, Cluster 0 comprises the most data points. This cluster may indicate the major group of typical customers. They usually visit some product related pages, spend the suitable amount of time browsing on these webs, and they leave pages at moderate rates, which means the average bounce rate and exit rate. Finally, Cluster (1) is the group of users who spend significant time on few pages. This cluster is much less than cluster.

KEY SUMMARY

Based on the observations of the pair relationships and machine learning models plotted, there are several key insights that online shops may modify accordingly to increase the revenue. Firstly, it is obvious that customers visit the Product_Related pages most frequently, so improving the quality of these pages is very crucial. We can optimise pages by creating attractive content, using appealing picture, providing detailed product descriptions, publishing the review and comments of previous buying, etc. This would attract the customers' attention and make them spend more time on Product pages, eventually leading to buy new products.

Secondly, since Browser 1 and Browser 2 take a high proportion of positive revenue, we need to make sure these browsers can function properly. In addition, we can improve the design, functionality, user experience of these platforms, so the users can have the best experience when browsing on. Thirdly, attracting new customers regularly is very important, since this group contributes more revenue than returning customers group and other groups. Shops can focus on various approaches, from marketing, advertising, social media, creating events to give discounts and vouchers, etc, to convert new customers into loyal ones. Finally, because low bounce and low exit rate can lead to higher revenue generation, maintaining low bounce rates and low exit rates can help increase the revenue. To do that, we can redesign the web to be more user-friendly, engaging instruction for new users and fast loading times, which keeps users spending more time browsing.

In the classification task, two models' performance assessed by evaluation metrics are shown in the table:

CLASSIFICATION	Accuracy	AUC score	Precision (Macro Avg)	Recall (Macro Avg)
Random Forest	0.874	0.837	0.76	0.84
Random Forest (Hyperparameter-tuning)	0.877	0.842	0.77	0.84
Stacking Model	0.88		0.77	0.84
Stacking Model (Hyperparameter-tuning)	1.0 (std = 0.0)	1.0 (std = 0.0)	1.0 (std = 0.0)	1.0 (std = 0.0)

Table 3: COMPARISION OF CLASSIFICATION MODELS

In the clustering task, two models' performance assessed by evaluation metrics are shown in the table below.

CLUSTERING	Silhouette Score
K_Means	0.497
DBSCAN	0.742

Table 4: COMPARISION OF CLUSTERING MODELS

CONCLUSION

By analyzing a rich dataset of customer web browsing behavior, this research project aimed to understand the factors that drive successful online purchases. The dataset, comprising a diverse range of customer attributes and website interaction metrics, presented a valuable opportunity to investigate the complex interplay between user characteristics, website engagement, and purchasing decisions. Through a combination of data exploration, feature engineering, and machine learning techniques, this research identified key predictors of customer conversion, providing valuable insights for businesses to optimize their online marketing strategies and improve customer experience. While there are limitations with our data and its application on a larger scale, our analysis shows that it is possible to predict a site visitor buying behavior within a certain level of confidence based on the features we were able to pull from the data set.

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